Section 3.4 The Chain Rule

① $f(x) = (7x+5)^2$

$f(x) = 49x^2 + 70x + 25$

$f'(x) = 98x + 70$

$f'(x) = 2(49x + 35)$

$f'(x) = 2(7x+5)\cdot 7$

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x))\cdot g'(x)$$

Ex ② $f(x) = 5\cdot[\sin x]^{5-1}$

$f'(x) = 5[\sin x]^4 \cdot \cos x$

Ex ③ $y = \sqrt{x^2+9}$ C.R.

$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

$$y' = \frac{1}{2\sqrt{x^2+9}} \cdot 2x$$

$$y' = \frac{x}{\sqrt{x^2+9}}$$

Ex ④ $y = \sec(e^{5x})$ C.R. C.R.

$$y' = \sec(e^{5x}) \cdot \tan(e^{5x}) \cdot e^{5x} \cdot 5$$

Ex ⑤ $f(x) = \frac{1}{x^3-x} = -1 \cdot (x^3-x)^{-1-1}$

$$f'(x) = -1 \cdot (x^3-x)^{-2} \cdot (3x^2-1)$$

$$f'(x) = \frac{-(3x^2-1)}{(x^3-x)^2}$$

Ex 6 $\quad g(x) = \left(\dfrac{1+x^2}{7x} \right)^4$

chain Rule

$$g'(x) = 4 \cdot \left(\frac{1+x^2}{7x} \right)^3 \cdot \left(\frac{2x(7x) - (1+x^2)\cdot 7}{(7x)^2} \right)$$

$$g'(x) = 4 \left(\frac{1+x^2}{7x} \right)^3 \left(\frac{7x^2 - 7}{49x^2} \right)$$

$$g'(x) = 4 \left(\frac{1+x^2}{7x} \right)^3 \frac{(x^2-1)}{7x^2}$$

EX 7 $y = \sin(\tan e^{x^2})$

$$y' = \cos(\tan e^{x^2}) \cdot \sec^2(e^{x^2}) \cdot e^{x^2} \cdot 2x$$

$$y = \sin x \cdot (\tan e^{x^2})$$

$$y' = \cos x \cdot \tan e^{x^2}$$

$$+ \sin x \cdot \sec^2(e^{x^2}) \cdot e^{x^2} \cdot 2x$$